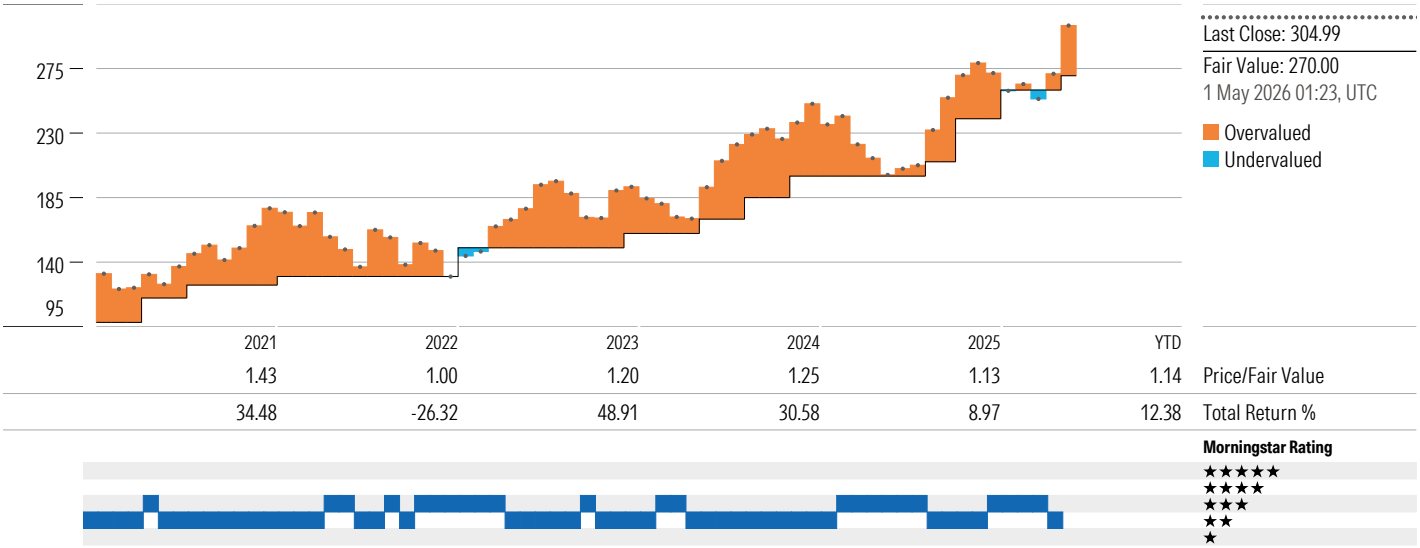
Apple Inc AAPL ★★ 22 May 2026 21:44, UTC

Last Price	Fair Value Estimate	Price/FVE	Market Cap	Economic Moat™	Equity Style Box	Uncertainty	Capital Allocation	ESG Risk Rating Assessment[1]
304.99 USD 21 May 2026	270.00 USD 1 May 2026 06:23, UTC	1.14	4.54 USD Tril 22 May 2026	Wide	Large Blend	Medium	Exemplary	22 May 2026 05:00, UTC

Price vs. Fair Value



	2021	2022	2023	2024	2025	YTD	
Price/Fair Value	1.43	1.00	1.20	1.25	1.13	1.14	
Total Return %	34.48	-26.32	48.91	30.58	8.97	12.38	

Total Return % as of 21 May 2026. Last Close as of 21 May 2026. Fair Value as of 1 May 2026 01:23, UTC.

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The primary analyst covering this company does not own its stock.

[1]The ESG Risk Rating Assessment is a representation of Sustainalytics' ESG Risk Rating.

Apple Earnings: Let the iPhone 17 Cycle Rip

Analyst Note William Kerwin, CFA, Senior Equity Analyst, 1 May 2026

Apple's March-quarter results beat the top ends of guidance. Revenue rose 17% year over year to $111 billion, led by iPhone revenue rising 22%. Gross margin of 49.3% was an all-time record. Management guided to similarly strong growth in the June quarter, with slight margin compression.

Why it matters: The iPhone's growth, led by the iPhone 17, continues to impress, particularly in China. We also appreciate strong profitability amid steep memory price inflation. We believe high growth will continue through the year, led by strong iPhone 17 uptake and a pent-up refresh cycle.

- The iPhone 17 cycle is the strongest since 2021, and it isn't driven by artificial intelligence. We forecast iPhone growth above 20% for the year, driven by what Apple does best: high-quality hardware (camera, display, processor), new form factors, and an ever-compelling software ecosystem.
- Against skyrocketing memory prices, we credit Apple for raising base iPhone storage capacities for a margin cushion and for excellent supply chain management through long-term contracts. Apple expects stronger memory cost headwinds in future quarters, but these remain a low impact to us.

The bottom line: We raise our fair value estimate for wide-moat Apple to $270 from $260, as we increase our 2026 growth forecast to reflect an even stronger iPhone 17 cycle. Shares rose 2% after hours on strong guidance, and we view the stock as fairly valued.

- Our forecast is led by iPhone as the majority of sales. After a superb 2026, we expect a return to mid-single-digit growth long term. We expect services to complement iPhone revenue, rising in the double

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Sector Technology

Industry Consumer Electronics

Business Description

Apple is among the largest companies in the world, with a broad portfolio of hardware and software products targeted at consumers and businesses. Apple's iPhone makes up a majority of the firm sales, and Apple's other products like Mac, iPad, and Watch are designed around the iPhone as the focal point of an expansive software ecosystem. Apple has progressively worked to add new applications, like streaming video, subscription bundles, and augmented reality. The firm designs its own software and semiconductors while working with subcontractors like Foxconn and TSMC to build its products and chips. Slightly less than half of Apple's sales come directly through its flagship stores, with a majority of sales coming indirectly through partnerships and distribution.

digits through 2030, led by Google's Search payments and App Store revenue.

- We model memory costs compressing margins by 100 basis points in the June quarter, and further in September. Still, gross margin of 48% is highly positive. We believe Apple can use long-term supply contracts to avoid the brunt of memory inflation, and high-margin services help pad profits.

Business Strategy & Outlook William Kerwin, CFA, Senior Equity Analyst, 1 May 2026

We believe Apple holds a dominant position at the forefront of the consumer electronics industry, focusing on a premium ecosystem of tightly integrated hardware, software, and services. We see the flagship iPhone as the linchpin of this ecosystem, from which Apple derives pricing power, switching costs, and a network effect. In our view, every other Apple device and service sees its greatest value from further locking in customers to this walled garden. This approach earns the firm a wide economic moat rating.

We are impressed with Apple's core design prowess, across both hardware and software, which we think is the product of immense cumulative research and development investments. We like Apple's push to bring most of its chip development in-house. To us, this gives the firm more opportunity for product customization and a better ability to differentiate. In our view, Apple reduces its own cyclicality compared with other consumer electronics players by melding its semiconductors, hardware, and software together.

Over the medium term, we expect Apple to focus on progressing its internal chip development, artificial intelligence capabilities, and further developing new form factors. Each one of these initiatives will hedge the firm against disruption risk, in our view. We also anticipate the company will continue returning tremendous amounts of cash back to shareholders, which is supported by its strong balance sheet.

We hold concerns over geopolitical and regulatory risk for Apple but don't see these threatening the firm's moat presently. Apple's supply chain is heavily concentrated in China and Taiwan, and disruptions in these regions, including tariffs, could limit its supply or profitability. Apple has thus far been adept at managing its complex supply chain and is actively diversifying into new regions, like India and the US. Apple has also been targeted by antitrust regulation, which threatens its profitability from services like the App Store. We don't anticipate severe rulings that would be detrimental to Apple's moat or core business, but we could see headwinds to its services profitability in the medium term.

Bulls Say William Kerwin, CFA, Senior Equity Analyst, 1 May 2026

- Apple offers an expansive ecosystem of tightly integrated hardware, software, and services, which locks in customers and generates strong profitability.
- We like Apple's move to in-house chip development, which we think has accelerated its product development and increased its differentiation.

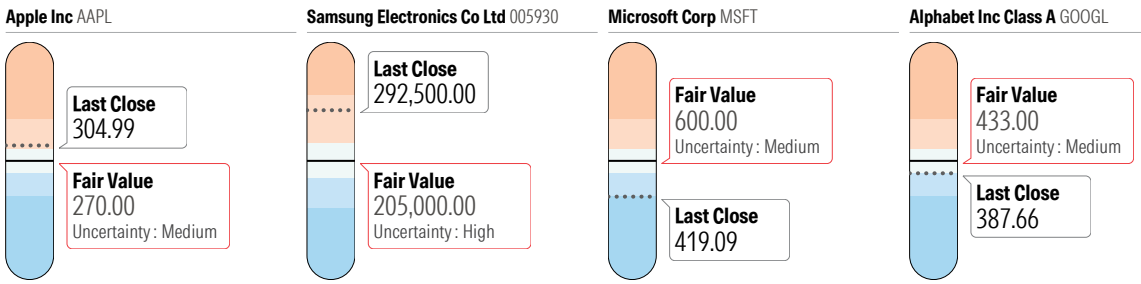
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Competitors



	Apple Inc AAPL	Samsung Electronics Co Ltd 005930	Microsoft Corp MSFT	Alphabet Inc Class A GOOGL
Economic Moat	Wide	None	Wide	Wide
Currency	USD	KRW	USD	USD
Fair Value	270.00 22 May 2026 16:44, UTC	205,000.00 22 May 2026 07:24, UTC	600.00 22 May 2026 16:43, UTC	433.00 22 May 2026 16:44, UTC
1-Star Price	364.50	317,750.00	810.00	584.55
5-Star Price	189.00	123,000.00	420.00	303.10
Assessment	Overvalued 22 May 2026	Overvalued 22 May 2026	Undervalued 22 May 2026	Undervalued 22 May 2026
Morningstar Rating	★★22 May 2026 05:00, UTC	★★22 May 2026 05:00, UTC	★★★★★22 May 2026 05:00, UTC	★★★★22 May 2026 05:00, UTC
Analyst	William Kerwin, Senior Equity Analyst	Jing Jie Yu, Equity Analyst	Dan Romanoff, Senior Equity Analyst	Malik Ahmed Khan, Senior Equity Analyst
Capital Allocation	Exemplary	Standard	Exemplary	Exemplary
Price/Fair Value	1.14	1.43	0.70	0.88
Price/Sales	10.15	4.42	9.81	11.07
Price/Book	42.54	4.08	7.50	9.69
Price/Earning	37.39	23.45	24.93	29.21
Dividend Yield	0.34%	0.57%	0.85%	0.22%
Market Cap	4,535.75 Bil	1,920,719.71 Bil	3,108.73 Bil	4,639.85 Bil
52-Week Range	193.46—311.40	53,700.00—300,500.00	356.28—555.45	162.00—408.61
Investment Style	Large Blend	Large Blend	Large Value	Large Blend

- Apple has a stellar balance sheet and sends great amounts of cash flow back to shareholders.

Bears Say William Kerwin, CFA, Senior Equity Analyst, 1 May 2026

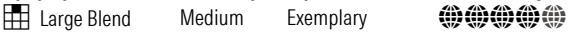
- Apple is prone to consumer spending and preferences, which creates cyclicality and opens the firm up to disruption.
- Apple's supply chain is highly concentrated in China and Taiwan, which opens up the firm to geopolitical risk. Attempts to diversify into other regions may pressure profitability or efficiency.
- Regulators have a keen eye on Apple, and recent regulations have chipped away at parts of Apple's sticky ecosystem.

Economic Moat William Kerwin, CFA, Senior Equity Analyst, 1 May 2026

We assign Apple a wide economic moat rating, stemming from customer switching costs, intangible assets, and a network effect. In our view, Apple's iOS ecosystem extends far-reaching, sticky tendrils

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into customers' wallets, entrenching customers with software capabilities and integration across disparate devices like the iPhone, Mac, iPad, Apple Watch, and more. We also see immense design prowess at Apple, most impressively from deep integration of hardware, software, and semiconductors to create best-of-breed products. Finally, we see a virtuous cycle between Apple's affluent customer base and vast ecosystem of developer partners. These moat sources elicit great profitability and returns on invested capital. In our view, Apple can leverage these moat sources into continued economic profits over the next 20 years, more likely than not.

Apple's most important moat source to us arises from the switching costs for its ecosystem of software, driven by iOS on the iPhone. Apple enjoys terrific customer retention and satisfaction, even despite pricing its products at a significant premium to its competition. First, Apple offers software capabilities that are only available to iPhone users: iMessage messaging and FaceTime calling, AirDrop sharing, Apple Pay digital wallet, and location sharing are among the most used. Apple's products become even more entrenched when a customer adopts two or more. Users that combine the iPhone with a Mac, an iPad, and/or an Apple Watch are offered more features that in turn create a higher cost to switching. Among these, we highlight iMessages shared across devices, health data shared securely between the Watch and iPhone, seamless transfer of media, copy and paste between devices, and complementary app interfaces across multiple devices.

In our view, Apple's ability to widen its portfolio of user devices helps augment its existing switching costs. The Apple Watch and AirPods are good recent examples of new products we see raising stickiness of customers. A Watch user can answer calls, read and respond to messages, and keep a tab on notifications, but must have an iPhone to use these capabilities. AirPods connect with marked ease to Apple devices but have to be manually re-paired each use if joined to a non-Apple device. As customers use more point devices, we believe they are less likely to leave Apple's ecosystem. We view Apple's nascent push into augmented and virtual reality (AR and VR) as the next step in this strategy. The firm's Vision Pro headset offers yet another auxiliary form factor that relies on the iPhone as the focal computing point.

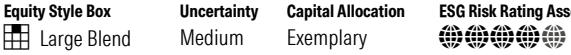
Furthermore, we see Apple's adept ability to introduce and succeed with new form factors (like smartwatches and VR headsets) as defending the firm from disruption risk. After Apple itself disrupted the mobile phone market by creating the modern smartphone, we believe it is critical for the firm to stay on the cutting edge and stave off disruptive threats to its own portfolio. We believe developing auxiliary products integrated with the iPhone is doing just that.

Beyond locking customers into a software ecosystem, we believe Apple's design expertise represents significant intangible assets that allow the firm to deliver top tier performance for its devices across hardware, software, and semiconductors. We see an enormous research and development budget totaling in the tens of billions of dollars enabling and maintaining this design prowess.

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Apple boasts impressive semiconductor design capabilities with which it augments the performance of its iPhone, Mac, iPad, and Watch offerings. We see chips as the highest-value component of consumer electronics. In designing its own chips, Apple is able to customize its specific devices for performance, power efficiency, and available features. Its A-series chips for the iPhone integrate neural engines for artificial intelligence, or AI, capabilities like facial recognition and are optimized for power efficiency to deliver best-in-class battery life. Its M-series chips for the Mac offer more memory than standardized PC chips from Intel or AMD, helping enable quicker compute and graphics performance. Owning its chip design also allows Apple to prioritize security. It embeds Secure Enclave in every Mac chip, which is a totally encrypted and physically separated part of the chip that protects biometric data and encryption keys.

By owning its chip design, Apple tightens the integration between its hardware devices and its sticky software ecosystem. Apple's custom chip families allow for common platforms across devices and new software features that further rope in customers. For example, Apple's focus on featuring significant neural engines on its chips allow for AI applications on devices, including facial recognition, voice assistance (Siri), live transcription, autocorrect, and highly adaptable software suggestions based on user preferences and history. We believe the common platform across chips for iPhone, Mac, iPad, and Watch also aid third-party app developers to create more integrated experiences for customers. Offering a better platform for developers leads to more apps designed for Apple devices, more features on third-party apps across devices, and a stickier proposition to customers, in our view.

Apple's chip capabilities are bolstered by a robust relationship with Taiwan Semiconductor, or TSMC, the largest foundry in the world. In our view, this relationship enables Apple to pursue more powerful and more efficient chips ahead of its competitors. Apple's iPhone is typically the first product to feature the newest process technology from TSMC, like the iPhone 15 Pro in 2023 being the first to feature 3-nanometer chips. Additionally, Apple pays TSMC for dedicated capacity, enabling Apple to be guaranteed a certain level of supply to meet demand for new iPhones, Macs, and the like.

We see Apple's chip design and software design prowess as impressive when taken separately but even more valuable in tandem. We believe the firm stands to benefit from quicker time-to-market for new features across both hardware and software by developing the two together and specified for its own devices. This compares with taking more general chips from the likes of Intel and being reliant on another firm's development for your own software ambitions (as Apple used to be for its Mac lineup before introducing the M-series chips). This integrated approach also stands out against Apple's primary smartphone competitor, Samsung, which owns its hardware assembly but outsources its chips and software to Qualcomm and Google (Android), respectively.

Apple focuses on the premium ends of its markets with best-of-breed devices that it sells at significant

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premiums. Customers have shown a consistent desire to pay up for this performance, generating unit market share gains for Apple even as its prices rise compared with competitors. Apple's share gains have been broad-based, across both developed and emerging markets (like China and India), consumers and enterprises, and every major product line, including iPhone, Mac, and iPad. We believe further share gains are in Apple's future, particularly as it sees the strongest preference among younger users.

Apple's moat is reinforced with a network effect between its large, committed user base and expansive ecosystem of application developers. We believe that more developers and more talented developers are drawn to Apple's large install base, while consumers are drawn to the wide selection of applications to which they gain access by purchasing an Apple device. In our view, this is accentuated by a richer base of Apple users, compared with those of competing PC or mobile vendors that spend more on apps and generate more developer profits. We also think that Apple's unified software ecosystem across iPhone, Mac, iPad, Watch, and more is an attractive proposition for developers. A developer can more easily gain multiple footholds in a user with tight integration and overlap between Apple's products and OSes, compared with designing disparate interfaces for an Android phone and Windows PC, for example. We think this network effect can be enhanced with more touchpoints in the Apple ecosystem, like the Vision Pro, that plug right into the existing development frameworks. We don't believe Apple's network precludes the existence of a network effect for a competing app store, like Google Play, but see Apple's rich network augmenting its user experience and supplementing its switching costs.

Apple's ability to deliver cutting-edge products with tightly integrated hardware and software, along with its software ecosystem's ability to lock in customers, results in impressive profit margins. On top of 45% gross margins, which are impressive for a consumer hardware provider, Apple's asset-light model with outsourced manufacturing generates robust returns on invested capital, to the tune of roughly 50%. In our view, Apple's sticky ecosystem and phenomenal engineering capabilities will enable it to continue earning strong economic profits over the next 20 years.

Fair Value and Profit Drivers William Kerwin, CFA, Senior Equity Analyst, 1 May 2026

Our fair value estimate for Apple is $270 per share. Our valuation implies a fiscal 2026 price/earnings multiple of 31 times, a fiscal 2026 enterprise value/revenue multiple of 8 times, and a fiscal 2026 free cash flow yield of 3%.

We project 9% compound annual revenue growth for Apple through fiscal 2030. The iPhone will be the most significant contributor to revenue over our forecast, and we project 9% growth in iPhone revenue over the next five years, including superb 20% growth in fiscal 2026, followed by mid-single-digit growth. We expect this to be driven primarily by unit sales growth, with modest price increases after a step up in fiscal 2026 as Apple raised its minimum iPhone storage capacities. We think longer-term

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pricing increases will be driven primarily by a mix shift toward the more premium Pro models. We attribute robust growth in fiscal 2026 to strong uptake of the iPhone 17 family, especially in China, and a pent-up refresh cycle from covid-era purchases five years prior.

Services are Apple's next biggest revenue contributor over our forecast, and we forecast more than 10% services revenue growth over the next five years. Services are driven in large part by revenue from Google for its status as the default search engine in Safari, as well as by Apple's cut of App Store revenue. We expect solid growth in Google revenue, but see a more mixed outlook for App Store results, where we forecast growth in overall app revenue but gradually lower cuts going to Apple later in the decade as a result of regulatory pressures to return more profits to developers. Elsewhere, we see mostly double-digit growth across revenue from Apple Music, Apple TV+, Apple Pay, AppleCare, and Apple's other services.

We see high-single-digit growth for Apple's wearables business through fiscal 2030. We see mid-single-digit growth for Apple Watch and AirPods revenue, with both products continuing to gain share. We expect double-digit growth for Apple's nascent Vision Pro, but expect this to remain a small minority of total revenue and a niche product in its current version. We project $2 billion in Vision Pro revenue in fiscal 2030.

Across Apple's other primary hardware products, Mac and iPad, we see midcycle growth in the low single digits. We expect Mac to see stronger short-term revenue from a cyclical rebound in consumer PC spending, as well as strong uptake for the lower-priced MacBook Neo.

We forecast gross margins to rise past 50% in fiscal 2030, up from 47% in fiscal 2025. We believe Apple can see margin expansion from a higher mix of higher-margin hardware, like iPhone Pro models, and services. We also expect Apple to continue using R&D to both maintain pricing power and trim costs. R&D helps Apple develop new features, especially by creating more cost-efficient semiconductors and bringing more chip development in-house. In the short-term, we see modest margin pressure from rising memory prices, but expect Apple to continue navigating these well and maintaining high margins. We believe pushing base-level iPhone storage capacities higher gives Apple a margin cushion, and see it adept at pressuring its supply chain to get supply on favorable terms.

Operating margins rise with gross margins in our forecast, reaching 36% in fiscal 2030 from 32% in fiscal 2025. We anticipate strong growth in research and development expenditures, which are key to Apple's value and moat. We observe robust R&D growth in fiscal 2026, which we think is ramping up Apple's internal investments in AI and AI-centric devices for the future. We think the firm can eke out modest leverage on other operating expenses, but anticipate higher short-term costs as Apple pivots some manufacturing out of China and invests in AI models.

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Risk and Uncertainty William Kerwin, CFA, Senior Equity Analyst, 1 May 2026

We assign Apple with a Medium Morningstar Uncertainty Rating. We see the firm's greatest risk as its reliance on consumer spending, for which there is great competition and cyclicality. Apple is at constant risk of disruption, just as the iPhone disrupted BlackBerry in the budding smartphone market. The iPhone could be unseated by a new device or "superapp." We view the firm defending against this risk, however, by introducing new form factors (like a watch and an augmented reality headset) and selling an ecosystem of software and services on top of hardware.

We also see geopolitical risk arising from Apple's supply chain. It is heavily dependent on Foxconn and Taiwan Semiconductor, or TSMC, for its assembly and chip production, respectively. The majority of iPhones are produced at a megafactory in China by Foxconn, and the majority of Apple chips are produced in Taiwan by TSMC. If there were a souring of relations between the US and China, or if China threatened Taiwan, Apple could see a severe hit to its supply. Additionally, the Chinese government has recommended government officials not conduct business on iPhones, which presents a current and potential future risk to Apple's revenue in China.

Apple also faces regulatory scrutiny. The European Union has enacted regulations forcing Apple to offer third-party app stores, adopt open messaging standards, and restricting Apple from gatekeeping its repair materials to third parties. Similarly, there are ongoing antitrust cases in the US that could affect Apple's walled garden ecosystem. All these help to chip away at the firm's differentiation, but we believe the firm is offsetting these by adding new stick points in customers, like new services and devices.

Finally, we see low environmental, social, and governance risk for Apple. The firm has committed to full carbon neutrality by 2030, and we believe it will achieve its goal. The potential future loss of talented human capital could be another risk on this front.

Capital Allocation William Kerwin, CFA, Senior Equity Analyst, 1 May 2026

We assign Apple an Exemplary Capital Allocation Rating, with Apple's investments as the primary driver to our rating. We see Apple's ability to develop updated and brand new technology internally as exceptional. Beyond iterations of products like the iPhone and Mac, Apple's internal chip and software development are particularly impressive to us. We believe Apple's semiconductors are strong, and the firm's ability to tailor chips to specific devices creates compelling value to customers. Apple's chips allow the firm to prioritize artificial intelligence applications and other proprietary software and features that generic chips from the likes of Intel may not.

We also believe Apple has invested in new products at the right time, seemingly perfecting the strategy of being a fast and premium follower. The Apple Watch is a prime example, and one where Apple now holds the largest share of its market. New products also create another touch point by which Apple can

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MORNINGSTAR®

Apple Inc AAPL ★★ 22 May 2026 21:44, UTC

Last Price	Fair Value Estimate	Price/FVE	Market Cap	Economic Moat™	Equity Style Box	Uncertainty	Capital Allocation	ESG Risk Rating Assessment[1]
304.99 USD 21 May 2026	270.00 USD 1 May 2026 06:23, UTC	1.14	4.54 USD Tril 22 May 2026	Wide	Large Blend	Medium	Exemplary	22 May 2026 05:00, UTC

lock in customers and raise its switching costs, reinforcing its wide moat. In our view, these also hedge against disruption risk by expanding Apple's ecosystem and creating prowess in new form factors that may eventually succeed the smartphone. We see the Vision Pro as the newest example of this strategy and one that will also leverage Apple's silicon expertise. We believe AI capabilities are the next frontier for Apple's investment. While we think the firm has thus far lagged the cutting edge here, we expect it to improve its effectiveness over time. For Apple, we don't see differentiation coming from building the best large language model, but in integrating useful features based on AI models. This is Apple's forte, and we expect it to bring more of these features to market in the short term.

On top of strong investments, Apple boasts a robust balance sheet and uses it to reward shareholders. Apple's cash generation is fantastic, and we like its goal of becoming net cash-neutral, which should help avoid the opportunity cost of holding significant amounts of cash. We also like the firm's dividend and buyback program, a combination that routinely sends all of Apple's free cash flow back to shareholders.

Underpinning Apple's capital allocation strategy, we are pleased with the firm's board of directors and Tim Cook as a chief executive. We believe Cook has been instrumental in pivoting Apple into an ecosystem company and has made savvy bets on new form factors, semiconductor design, and services. We like his compensation structure that offers a relatively low, flat salary with the majority of his pay coming from incentive-based compensation. Apple's board is made up of blue-chip current and former executives across industries. The board has good expertise across finance, biotechnology, and consumer products, which likely informs Apple's strategy in existing and new markets.

Analyst Notes Archive

Apple: Passing the Big Baton William Kerwin, CFA,Senior Equity Analyst,21 Apr 2026

Apple named John Ternus its next CEO, effective Sept. 1. Tim Cook will transition to the role of executive chairman, and former board chair Arthur Levinson will assume the role of lead independent director. Ternus was formerly the senior vice president of hardware engineering. Why it matters: We expect continuity in Apple's strategy under Ternus. We like that the new CEO will have a background in products and engineering, and expect a continued firmwide focus on the user experience, security and privacy, and deeper integration with Apple-designed silicon. Ternus, 50 years old, had been rumored to be the internal successor for several months, with Cook turning 65 in 2025 after 15 years in charge. Ternus has been credited with hardware design on the Mac and iPad, working in hardware and products for 25 years at Apple. Apple's artificial intelligence strategy will be Ternus' first focus. We believe Apple is ready to release improved AI software under its partnership with Google Gemini this year, and we expect Ternus to focus on the integration of distilled Google models with Apple's own software and hardware ecosystem. The bottom line: We maintain our $260 fair value estimate for wide-moat Apple and see shares as fairly valued. Ternus was our expected successor to Cook, and we expect

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continuity in a transition. Shares were effectively flat after hours on the news. Coming up: Apple reports March-quarter earnings after the market closes on Thursday, April 30. We expect strong year-over-year growth to be led by the iPhone, and another record quarter for gross margin, in spite of rising memory prices. We expect to hear more from Ternus and Cook on the earnings call. Apple's worldwide developer conference, WWDC, is set for June 8 through June 12. There, we expect the announcement of revamped AI features under Apple Intelligence, and for Ternus to participate meaningfully in the keynote, which we expect to be on June 8.

Apple: March Releases, Including Brand-New MacBook Neo, Keep Pricing Low Despite Memory Inflation William Kerwin, CFA,Senior Equity Analyst,4 Mar 2026

Apple unveiled a slew of new products during the week of March 2, highlighted by the brand-new MacBook Neo. Other releases included updated iterations, such as the iPhone 17e, new M5-series chips in MacBooks, a new iPad Air with an updated M4 chip, and new Studio displays. Why it matters: For the most part, these updates were evolutionary. We note a heavy emphasis on AI-readiness across all product releases, and that Apple refrained from raising prices across the lineup. We remain impressed by Apple's product and silicon ecosystem, with tight software integration. Despite skyrocketing memory chip prices, Apple kept prices constant across Mac and iPad, consistent with the iPhone 17 family. We like Apple absorbing memory prices, and we believe the firm is adept at negotiating a ballooning pricing environment while maintaining a strong margin. New products include updated processors, along with connectivity chips like N1 (Wi-Fi and Bluetooth) and C1X (cellular). Vertically integrated silicon design helps Apple keep prices down, raise margins, and develop more tightly integrated software and hardware. The bottom line: We maintain our $260 fair value estimate for wide-moat Apple, as evolutionary updates align with our pricing expectations and reinforce our view of Apple's strong product lineup. Shares look fairly valued and didn't move much on releases. We expect Apple to continue expanding margins, even amid tariffs and memory cost pressures. We believe Apple's premium prices have built-in buffers for component cost inflation and that it receives best-in-class contract pricing for memory, well below astronomical spot prices. Apple has lowered inflation-indexed prices by close to 20% since 2020, while expanding gross margin by more than 800 basis points. We credit best-in-class supply chain management and continued vertical integration for Apple's margin performance.

Apple Earnings: Prolific Quarter Portends Terrific 2026 Growth William Kerwin, CFA,Senior Equity Analyst,30 Jan 2026

Apple posted stellar December-quarter results, led by 23% year-over-year growth in iPhone revenue and over 100 basis points of gross margin expansion. March-quarter guidance implies continued high growth and further margin expansion. Why it matters: IPhone growth was stunning, coming in about 10% higher than our model. We believe Apple is benefiting from a strong refresh cycle in 2026, driven

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by pent-up customer demand. Profitability also impressed us amid headwinds from memory prices and ongoing tariff costs. Revenue growth for China was an exceptional 38% year over year and shows a sharp inflection from two years of underperformance. We still see longer-term headwinds to Apple's China growth, but this quarter shows the firm's ability to compete and win against beefed-up domestic competition. Supply constraints are limiting Apple's ability to meet demand, primarily for iPhone and AirPods. We expect limited supply, predominantly as Apple competes for chip production capacity at TSMC with the likes of Nvidia, to endure through the year but to have a low impact on results. The bottom line: We raise our fair value estimate for wide-moat Apple to $260 per share from $240, reflecting stronger short-term iPhone growth and higher profitability. Shares were slightly up after hours and appear fairly valued to us. We now expect a strong iPhone growth cycle in fiscal 2026, with revenue growth in the low teens, up from high single digits previously. We continue to see mid-single-digit growth over the long term, but Apple is benefiting from a strong refresh cycle with the iPhone 17 family in the short term. Apple's relentless gross margin expansion, despite tariffs and skyrocketing memory chip prices, is raising our expectations. We project Apple will reach a 50% gross margin in two years, up from 38% in 2020. We attribute this expansion to Apple silicon and expanding services sales.

Apple: Gemini Partnership Lets AI-Worried Apple Investors Breathe a Sigh of Relief William Kerwin, CFA,Senior Equity Analyst,12 Jan 2026

On Jan. 12, Google announced that Apple's next-generation artificial intelligence models will be based on Gemini models in a multiyear partnership. Gemini will now be a foundational piece of Apple Intelligence and will underpin the next-generation Siri voice assistant set to launch in 2026. Why it matters: This is the catalyst Apple needs to revamp its AI strategy, which has promised much and delivered little so far. Gemini should allow Apple's vision to bear out. We believe a competitive—if not leading—AI feature suite will reinforce Apple's wide-moat mobile device ecosystem. We like Apple's choice to partner, rather than build generative AI models from the ground up. Apple's strength is building software applications users love. To us, Apple doesn't need to develop frontier models but rather should use a strong model like Gemini to undergird compelling features. Apple's brand of security and privacy should remain intact, as it will use Gemini instances on its own servers in its own data centers via its Private Cloud Compute offering for AI processing. We expect users will be able to opt in to sharing prompts with Gemini directly, as well. The bottom line: We maintain our $240 fair value estimate for wide-moat Apple, as the announcement with Gemini was an expected part of our long-term thesis. Shares are fairly valued. The stock didn't move much on the announcement, implying the market already priced in the partnership. To us, this will protect Apple's ecosystem and existing growth trajectory, rather than spurring a growth cycle. To drive a higher valuation, we would need to see Gemini spur a hardware refresh cycle for the iPhone. Bulls say: If the Gemini partnership quickly enables new, larger models for Apple that require greater memory to run compelling features, we could see a stronger upcycle. This would entail consumers buying the newest phones to be able to use new,

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AI-generated features.

Apple Earnings: Strong Performance Overpowers Headwinds, and We Raise Our Valuation by 14% to $240 William Kerwin, CFA,Senior Equity Analyst,31 Oct 2025

Apple's September-quarter results were strong, with revenue rising 8% year over year to $102 billion and gross margin expanding 100 basis points year over year to 47.2%. December guidance was even better, with double-digit revenue growth expected, as well as further gross margin expansion. Why it matters: Exceptional iPhone revenue guidance of double-digit year-over-year growth for the next quarter well exceeded our model, and demonstrates strong uptake of the iPhone 17 family, including the new iPhone Air. We like that Apple is seeing strong growth even against headwinds out of China. Services revenue rose a strong 15% year over year, and guidance implies continuing momentum. Apple's services business looks more secure after an antitrust remedy ruling for Google in September allowed the firm to continue paying Apple to be the default browser on its devices. Profitability continues to shine, even as Apple eats up additional tariff costs. We assess tariffs as a roughly 100-basis-point margin headwind to Apple, which makes its guidance to record gross margin in the December quarter all the more impressive. The bottom line: We raise our fair value estimate for wide-moat Apple to $240 per share, from $210. We're impressed by growth and profitability despite a slow artificial intelligence feature rollout, tariffs, and China headwinds. Still, we find the market's valuation challenging. iPhone and services drive firmwide growth, and we expect a good year for iPhone revenue after a few years of lower growth. We expect services to continue growing in the low teens, especially with more conviction in the durability of payments from Google. We expect gross margins to continue marching upward, as Apple benefits from a higher mix of high-margin services revenue and commands pricing power in premium iPhone models. We also expect in-house chip development to continue enabling lower marginal product costs.

Apple: Not Much More Air Under Its Wings With Mostly Evolutionary iPhone Updates William Kerwin, CFA,Senior Equity Analyst,9 Sep 2025

Apple unveiled its new iPhone 17 lineup on Tuesday, including a brand new, ultrathin iPhone Air. Management also announced updates to the Watch lineup, including Series 11 and Ultra 3, and the AirPods Pro 3. Why it matters: The iPhone 17 family met our expectations and shows a continuing evolution of Apple's flagship product. We find the new Air model attractive, but we still don't expect the new form factor to spur a significant unit sales growth inflection for the firm. Apple emphasized its existing artificial intelligence feature set throughout, including Live Translation and Visual Intelligence features. We like Apple's AI features, but we view them as incremental. We look for a bigger overhaul of Siri and a potential partnership with Google Gemini to spur an inflection in unit sales. We continue to see Apple's hardware and software design as second to none. We see the iPhone 17 family as largely evolutionary, but reinforcing Apple's compelling iOS ecosystem that drives its wide economic moat. The

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bottom line: We maintain our $210 fair value estimate for wide-moat Apple. We continue to see longer-term iPhone unit sales growth slowing, and don't see the new Air model as a positive catalyst. Shares cut back by 2% intraday on the releases, and look moderately overvalued to us. We model 5% annualized growth for iPhone revenue through fiscal 2029, compared with a 10-year historical growth rate of 7%. This is driven by our expectations of continued growth headwinds from China and slowing consumer refresh cycles globally. Coming up: All of the new products launch on Sept. 19. We expect the latest releases to drive typical seasonal revenue growth for Apple. We expect a slight sequential uptick in the September quarter with 12 days of new-product sales, and significant sequential growth in the December quarter.

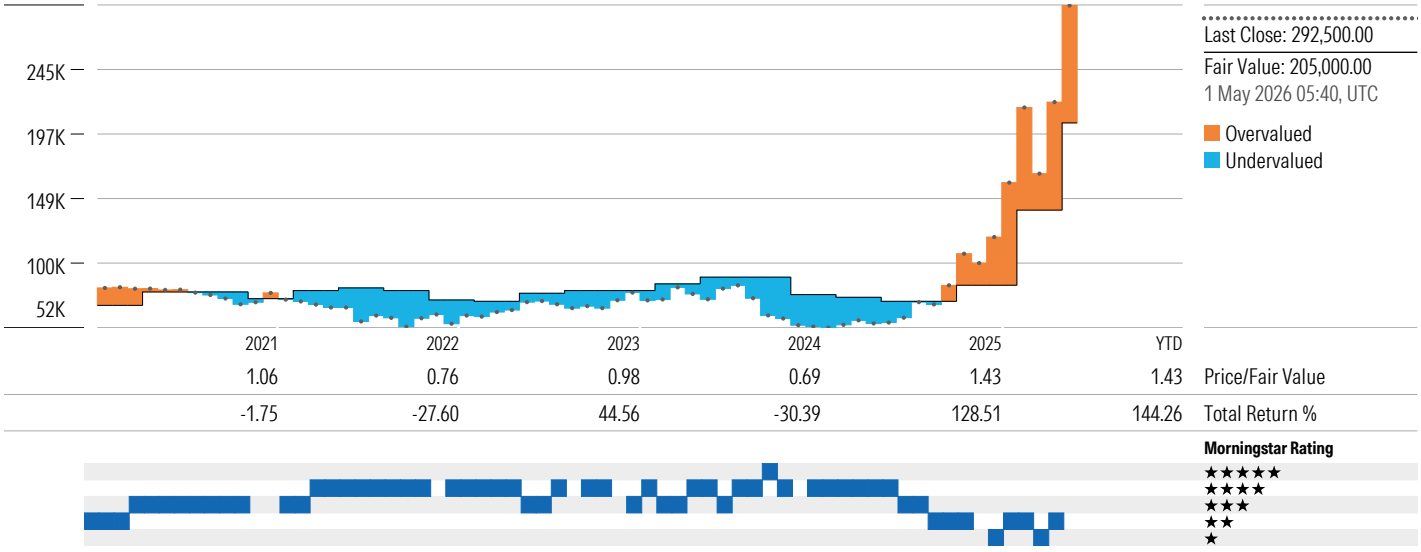
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Competitors Price vs. Fair Value

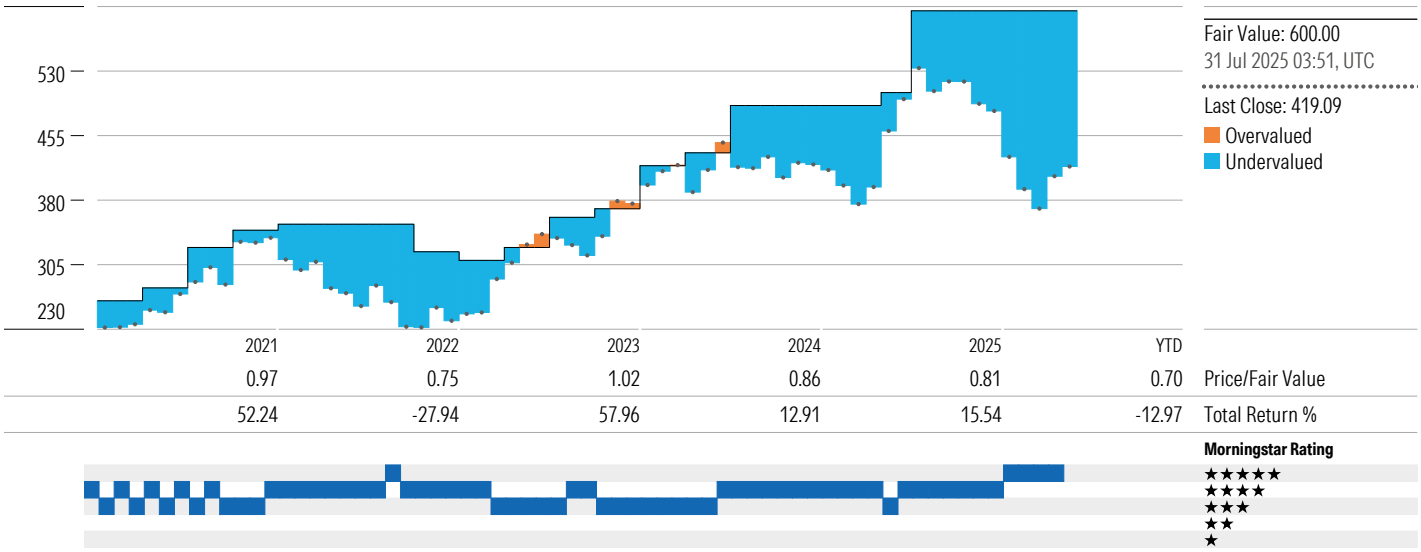
Samsung Electronics Co Ltd 005930



	2021	2022	2023	2024	2025	YTD	
	1.06	0.76	0.98	0.69	1.43	1.43	Price/Fair Value
	-1.75	-27.60	44.56	-30.39	128.51	144.26	Total Return %

Total Return % as of 22 May 2026. Last Close as of 22 May 2026. Fair Value as of 1 May 2026 05:40, UTC.

Microsoft Corp MSFT



	2021	2022	2023	2024	2025	YTD	
	0.97	0.75	1.02	0.86	0.81	0.70	Price/Fair Value
	52.24	-27.94	57.96	12.91	15.54	-12.97	Total Return %

Total Return % as of 21 May 2026. Last Close as of 21 May 2026. Fair Value as of 31 Jul 2025 03:51, UTC.

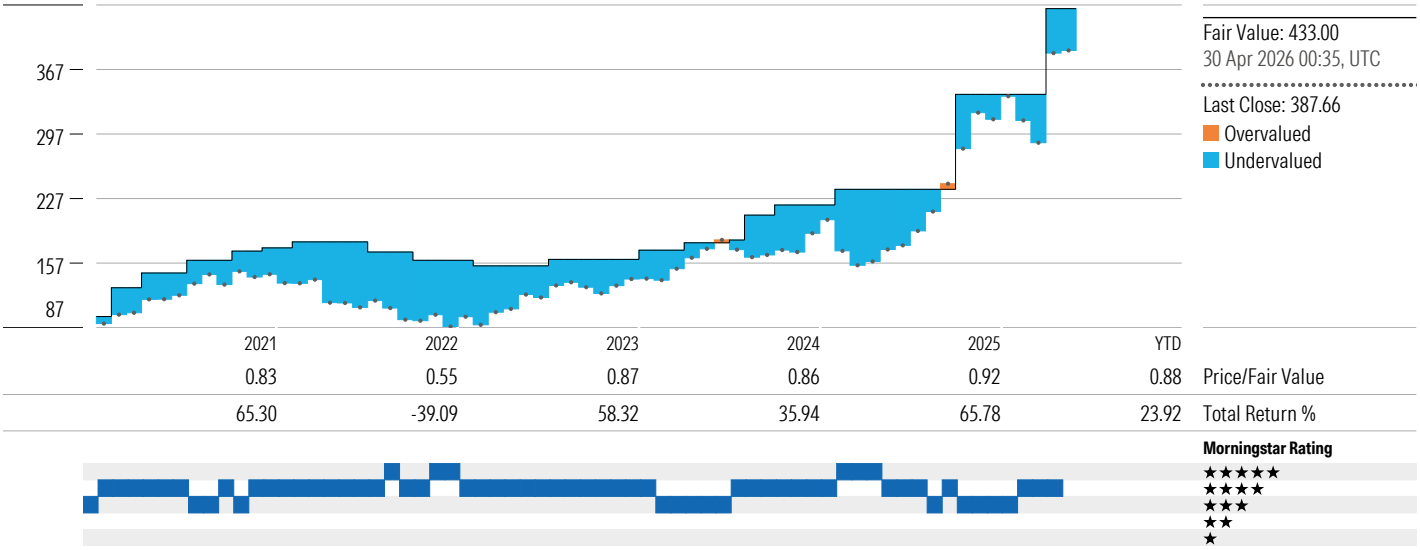
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Competitors Price vs. Fair Value

Alphabet Inc Class A GOOGL



	2021	2022	2023	2024	2025	YTD	
	0.83	0.55	0.87	0.86	0.92	0.88	Price/Fair Value
	65.30	-39.09	58.32	35.94	65.78	23.92	Total Return %

Total Return % as of 21 May 2026. Last Close as of 21 May 2026. Fair Value as of 30 Apr 2026 00:35, UTC.

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Morningstar Valuation Model Summary

Financials as of 30 Apr 2026	Actual			Forecast				
Fiscal Year, ends 30 Sep	**2023**	**2024**	**2025**	**2026**	**2027**	**2028**	**2029**	**2030**
Revenue (USD Mil)	383,285	391,035	416,161	482,231	521,113	556,520	592,863	631,828
Operating Income (USD Mil)	114,301	123,216	133,050	157,330	174,557	191,170	208,684	227,812
EBITDA (USD Mil)	125,820	134,661	144,748	170,785	189,017	206,665	225,339	245,792
Adjusted EBITDA (USD Mil)	125,820	134,661	144,748	170,785	189,017	206,665	225,339	245,792
Net Income (USD Mil)	96,995	93,736	112,010	130,120	144,883	158,671	173,208	189,084
Adjusted Net Income (USD Mil)	96,995	103,982	112,010	130,120	144,883	158,671	173,208	189,084
Free Cash Flow To The Firm (USD Mil)	95,760	119,719	93,621	114,424	143,217	156,432	171,237	187,195
Weighted Average Diluted Shares Outstanding (Bil)	16	15	15	15	14	14	14	13
Earnings Per Share (Diluted) (USD)	6.13	6.08	7.47	8.84	10.07	11.34	12.78	14.39
Adjusted Earnings Per Share (Diluted) (USD)	6.13	6.75	7.47	8.84	10.07	11.34	12.78	14.39
Dividends Per Share (USD)	0.94	0.98	1.02	1.06	1.12	1.20	1.28	1.36

Margins & Returns as of 30 Apr 2026		Actual			Forecast					
	3 Year Avg	**2023**	**2024**	**2025**	**2026**	**2027**	**2028**	**2029**	**2030**	**5 Year Avg**
Operating Margin %	31.1	29.8	31.5	32.0	32.6	33.5	34.4	35.2	36.1	34.4
EBITDA Margin %	—	32.8	34.4	34.8	35.4	36.3	37.1	38.0	38.9	—
Adjusted EBITDA Margin %	—	32.8	34.4	34.8	35.4	36.3	37.1	38.0	38.9	37.2
Net Margin %	25.4	25.3	24.0	26.9	27.0	27.8	28.5	29.2	29.9	28.5
Adjusted Net Margin %	26.3	25.3	26.6	26.9	27.0	27.8	28.5	29.2	29.9	28.5
Free Cash Flow To The Firm Margin %	26.0	25.0	30.6	22.5	23.7	27.5	28.1	28.9	29.6	27.6

Growth & Ratios as of 30 Apr 2026		Actual			Forecast					
	3 Year CAGR	**2023**	**2024**	**2025**	**2026**	**2027**	**2028**	**2029**	**2030**	**5 Year CAGR**
Revenue Growth %	1.8	-2.8	2.0	6.4	15.9	8.1	6.8	6.5	6.6	8.7
Operating Income Growth %	3.7	-4.3	7.8	8.0	18.3	11.0	9.5	9.2	9.2	11.4
EBITDA Growth %	3.6	-3.6	7.0	7.5	18.0	10.7	9.3	9.0	9.1	11.2
Adjusted EBITDA Growth %	3.5	-3.6	7.0	7.5	18.0	10.7	9.3	9.0	9.1	11.2
Earnings Per Share Growth %	6.9	0.3	-0.8	22.7	18.4	13.9	12.7	12.6	12.7	14.0
Adjusted Earnings Per Share Growth %	6.9	0.3	10.0	10.6	18.4	13.9	12.7	12.6	12.7	14.0

Valuation as of 30 Apr 2026	Actual			Forecast				
	2023	**2024**	**2025**	**2026**	**2027**	**2028**	**2029**	**2030**
Price/Earning	27.9	34.5	34.1	34.9	30.7	27.2	24.2	21.5
Price/Sales	7.0	9.0	9.0	9.4	8.7	8.2	7.7	7.2
Price/Book	43.6	63.0	51.9	42.0	35.4	35.7	33.3	30.9
Price/Cash Flow	—	—	—	—	—	—	—	—
EV/EBITDA	21.6	26.4	26.3	26.7	24.1	22.0	20.2	18.5
EV/EBIT	23.8	28.9	28.6	28.9	26.1	23.8	21.8	20.0
Dividend Yield %	0.5	0.4	0.4	0.3	0.4	0.4	0.4	0.4
Dividend Payout %	15.3	14.5	13.7	12.0	11.1	10.6	10.0	9.5
Free Cash Flow Yield %	—	—	—	—	—	—	—	—

Operating Performance / Profitability as of 30 Apr 2026	Actual			Forecast				
Fiscal Year, ends 30 Sep	**2023**	**2024**	**2025**	**2026**	**2027**	**2028**	**2029**	**2030**
ROA %	27.5	25.7	31.2	33.9	36.1	38.5	41.2	44.0
ROE %	156.1	164.6	151.9	120.2	115.5	131.2	137.9	143.8
ROIC %	89.8	86.6	81.8	82.9	78.2	73.8	71.9	71.2

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Apple Inc AAPL ★★ 22 May 2026 21:44, UTC

Last Price	Fair Value Estimate	Price/FVE	Market Cap	Economic Moat™	Equity Style Box	Uncertainty	Capital Allocation	ESG Risk Rating Assessment[1]
304.99 USD 21 May 2026	270.00 USD 1 May 2026 06:23, UTC	1.14	4.54 USD Tril 22 May 2026	Wide	Large Blend	Medium	Exemplary	22 May 2026 05:00, UTC

Financial Leverage (Reporting Currency)	Actual			Forecast				
Fiscal Year, ends 30 Sep	**2023**	**2024**	**2025**	**2026**	**2027**	**2028**	**2029**	**2030**
Debt/Capital %	4.0	2.9	2.5	2.0	1.8	1.9	1.8	1.7
Assets/Equity	5.7	6.4	4.9	3.5	3.2	3.4	3.3	3.3
Net Debt/EBITDA	0.4	0.3	0.3	0.3	0.1	0.1	0.1	0.0
Total Debt/EBITDA	0.9	0.8	0.7	0.5	0.4	0.4	0.4	0.4
EBITDA/ Net Interest Expense	222.7	-500.6	450.9	-490.8	—	—	—	—

Forecast Revisions as of 30 Apr 2026	2026		2027		2028	
Prior data as of 29 Jan 2026	**Current**	**Prior**	**Current**	**Prior**	**Current**	**Prior**
Fair Value Estimate Change (Trading Currency)	270.00	259.56	—	—	—	—
Revenue (USD Mil)	482,231	463,954	521,113	497,462	556,520	531,606
Operating Income (USD Mil)	157,330	152,873	174,557	168,464	191,170	182,971
EBITDA (USD Mil)	170,785	165,944	189,017	182,404	206,665	197,894
Net Income (USD Mil)	130,120	126,347	144,883	141,510	158,671	153,696
Earnings Per Share (Diluted) (USD)	8.84	8.61	10.07	9.90	11.34	11.09
Adjusted Earnings Per Share (Diluted) (USD)	8.84	8.61	10.07	9.90	11.34	11.09
Dividends Per Share (USD)	1.06	1.06	1.12	1.12	1.20	1.20

Key Valuation Drivers as of 30 Apr 2026

Cost of Equity %	9.2
Pre-Tax Cost of Debt %	5.5
Weighted Average Cost of Capital %	9.1
Long-Run Tax Rate %	—
Stage II EBI Growth Rate %	9.0
Stage II Investment Rate %	2.0
Perpetuity Year	20

Additional estimates and scenarios available for download at https://pitchbook.com/.

Discounted Cash Flow Valuation as of 30 Apr 2026

	USD Mil
Present Value Stage I	507,789
Present Value Stage II	1,814,370
Present Value Stage III	1,476,739
Total Firm Value	**3,798,899**
Cash and Equivalents	35,934
Debt	90,678
Other Adjustments	0
Equity Value	**3,744,155**
Projected Diluted Shares	15
Fair Value per Share (USD)	**270.00**

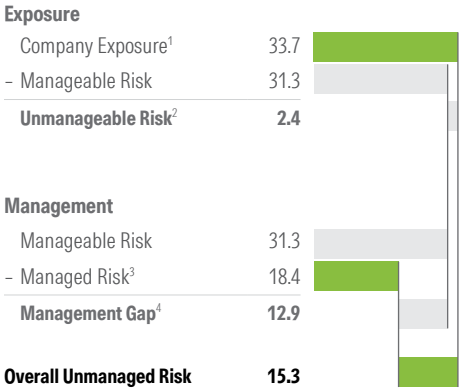
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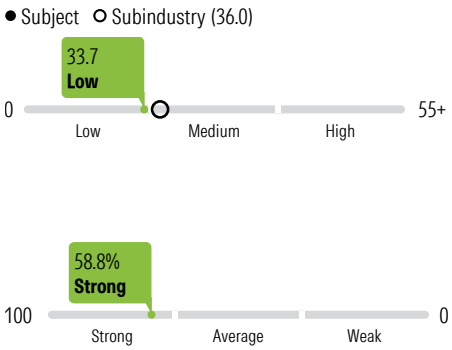
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Apple Inc AAPL ★★ 22 May 2026 21:44, UTC

Last Price	Fair Value Estimate	Price/FVE	Market Cap	Economic Moat™	Equity Style Box	Uncertainty	Capital Allocation	ESG Risk Rating Assessment[1]
304.99 USD 21 May 2026	270.00 USD 1 May 2026 06:23, UTC	1.14	4.54 USD Tril 22 May 2026	Wide	Large Blend	Medium	Exemplary	 22 May 2026 05:00, UTC

ESG Risk Rating Breakdown





- Exposure represents a company's vulnerability to ESG risks driven by their business model
- Exposure is assessed at the Subindustry level and then specified at the company level
- Scoring ranges from 0-55+ with categories of low, medium, and high-risk exposure

- Management measures a company's ability to manage ESG risks through its commitments and actions
- Management assesses a company's efficiency on ESG programs, practices, and policies
- Management score ranges from 0-100% showing how much manageable risk a company is managing

ESG Risk Rating



ESG Risk Ratings measure the degree to which a company's value is impacted by environmental, social, and governance risks, by evaluating the company's ability to manage the ESG risks it faces.

1. A company's Exposure to material ESG issues 2. Unmanageable Risk refers to risks that are inherent to a particular business model that cannot be managed by programs or initiatives 3. Managed Risk = Manageable Risk multiplied by a Management score of 58.8% 4. Management Gap assesses risks that are not managed, but are considered manageable 5. ESG Risk Rating Assessment = Overall Unmanaged Risk = Management Gap plus Unmanageable Risk

ESG Risk Rating Assessment[5]



ESG Risk Rating is of May 22, 2026. Highest Controversy Level is as of May 08, 2026. Sustainalytics Subindustry: Technology Hardware. Sustainalytics provides Morningstar with company ESG ratings and metrics on a monthly basis and as such, the ratings in Morningstar may not necessarily reflect current Sustainalytics' scores for the company. For the most up to date rating and more information, please visit: sustainalytics.com/esg-ratings/.

Peer Analysis 22 May 2026

Peers are selected from the company's Sustainalytics-defined Subindustry and are displayed based on the closest market cap values

Company Name	Exposure			Management			ESG Risk Rating		
Apple Inc	33.7 | Low	0	55+	58.8 | Strong	100	0	15.3 | Low	0	40+
Alphabet Inc	42.6 | Medium	0	55+	57.3 | Strong	100	0	19.9 | Low	0	40+
Microsoft Corp	32.8 | Low	0	55+	60.3 | Strong	100	0	14.4 | Low	0	40+
Samsung Electronics Co Ltd	37.5 | Medium	0	55+	71.5 | Strong	100	0	12.6 | Low	0	40+
Broadcom Inc	42.7 | Medium	0	55+	57.1 | Strong	100	0	20.1 | Medium	0	40+

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Appendix

Historical Morningstar Rating

Apple Inc AAPL 22 May 2026 05:00, UTC

Dec 2026	Nov 2026	Oct 2026	Sep 2026	Aug 2026	Jul 2026	Jun 2026	May 2026	Apr 2026	Mar 2026	Feb 2026	Jan 2026
—	—	—	—	—	—	—	★★	★★★	★★★	★★★	★★★
Dec 2025	Nov 2025	Oct 2025	Sep 2025	Aug 2025	Jul 2025	Jun 2025	May 2025	Apr 2025	Mar 2025	Feb 2025	Jan 2025
★★	★★	★★	★★	★★★	★★★	★★★	★★★	★★★	★★★	★★	★★
Dec 2024	Nov 2024	Oct 2024	Sep 2024	Aug 2024	Jul 2024	Jun 2024	May 2024	Apr 2024	Mar 2024	Feb 2024	Jan 2024
★★	★★	★★	★★	★★	★★	★★	★★	★★★	★★★	★★	★★
Dec 2023	Nov 2023	Oct 2023	Sep 2023	Aug 2023	Jul 2023	Jun 2023	May 2023	Apr 2023	Mar 2023	Feb 2023	Jan 2023
★★	★★	★★★	★★	★★	★★	★★	★★	★★★	★★★	★★★	★★★
Dec 2022	Nov 2022	Oct 2022	Sep 2022	Aug 2022	Jul 2022	Jun 2022	May 2022	Apr 2022	Mar 2022	Feb 2022	Jan 2022
★★★	★★★	★★	★★★	★★	★★	★★★	★★★	★★	★★	★★	★★
Dec 2021	Nov 2021	Oct 2021	Sep 2021	Aug 2021	Jul 2021	Jun 2021	May 2021	Apr 2021	Mar 2021	Feb 2021	Jan 2021
★★	★★	★★	★★	★★	★★	★★	★★★	★★	★★	★★	★★

Samsung Electronics Co Ltd 005930 22 May 2026 05:00, UTC

Dec 2026	Nov 2026	Oct 2026	Sep 2026	Aug 2026	Jul 2026	Jun 2026	May 2026	Apr 2026	Mar 2026	Feb 2026	Jan 2026
—	—	—	—	—	—	—	★★	★	★★	★★	★
Dec 2025	Nov 2025	Oct 2025	Sep 2025	Aug 2025	Jul 2025	Jun 2025	May 2025	Apr 2025	Mar 2025	Feb 2025	Jan 2025
—	★★	★★	★★	★★★	★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★
Dec 2024	Nov 2024	Oct 2024	Sep 2024	Aug 2024	Jul 2024	Jun 2024	May 2024	Apr 2024	Mar 2024	Feb 2024	Jan 2024
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Dec 2023	Nov 2023	Oct 2023	Sep 2023	Aug 2023	Jul 2023	Jun 2023	May 2023	Apr 2023	Mar 2023	Feb 2023	Jan 2023
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Dec 2022	Nov 2022	Oct 2022	Sep 2022	Aug 2022	Jul 2022	Jun 2022	May 2022	Apr 2022	Mar 2022	Feb 2022	Jan 2022
—	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★	★★★	—
Dec 2021	Nov 2021	Oct 2021	Sep 2021	Aug 2021	Jul 2021	Jun 2021	May 2021	Apr 2021	Mar 2021	Feb 2021	Jan 2021
—	★★★	★★★	★★★	★★★	★★★	★★★	★★★	★★★	★★	★★	★★

Microsoft Corp MSFT 22 May 2026 05:00, UTC

Dec 2026	Nov 2026	Oct 2026	Sep 2026	Aug 2026	Jul 2026	Jun 2026	May 2026	Apr 2026	Mar 2026	Feb 2026	Jan 2026
—	—	—	—	—	—	—	★★★★★	★★★★★	★★★★★	★★★★★	★★★★
Dec 2025	Nov 2025	Oct 2025	Sep 2025	Aug 2025	Jul 2025	Jun 2025	May 2025	Apr 2025	Mar 2025	Feb 2025	Jan 2025
★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★	★★★★	★★★★	★★★★	★★★★	★★★★
Dec 2024	Nov 2024	Oct 2024	Sep 2024	Aug 2024	Jul 2024	Jun 2024	May 2024	Apr 2024	Mar 2024	Feb 2024	Jan 2024
★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★	★★★	★★★	★★★	★★★	★★★
Dec 2023	Nov 2023	Oct 2023	Sep 2023	Aug 2023	Jul 2023	Jun 2023	May 2023	Apr 2023	Mar 2023	Feb 2023	Jan 2023
★★★	★★★	★★★★	★★★★	★★★	★★★	★★★	★★★	★★★	★★★★	★★★★	★★★★
Dec 2022	Nov 2022	Oct 2022	Sep 2022	Aug 2022	Jul 2022	Jun 2022	May 2022	Apr 2022	Mar 2022	Feb 2022	Jan 2022
★★★★	★★★★	★★★★	★★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★
Dec 2021	Nov 2021	Oct 2021	Sep 2021	Aug 2021	Jul 2021	Jun 2021	May 2021	Apr 2021	Mar 2021	Feb 2021	Jan 2021
★★★	★★★	★★★	★★★★	★★★	★★★★	★★★	★★★★	★★★	★★★★	★★★	★★★★

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Alphabet Inc Class A GOOGL 22 May 2026 05:00, UTC

Dec 2026	Nov 2026	Oct 2026	Sep 2026	Aug 2026	Jul 2026	Jun 2026	May 2026	Apr 2026	Mar 2026	Feb 2026	Jan 2026
—	—	—	—	—	—	—	★★★★	★★★★	★★★★	★★★	★★★
Dec 2025	Nov 2025	Oct 2025	Sep 2025	Aug 2025	Jul 2025	Jun 2025	May 2025	Apr 2025	Mar 2025	Feb 2025	Jan 2025
★★★	★★★	★★★★	★★★	★★★★	★★★★	★★★★	★★★★★	★★★★★	★★★★★	★★★★	★★★★
Dec 2024	Nov 2024	Oct 2024	Sep 2024	Aug 2024	Jul 2024	Jun 2024	May 2024	Apr 2024	Mar 2024	Feb 2024	Jan 2024
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Dec 2023	Nov 2023	Oct 2023	Sep 2023	Aug 2023	Jul 2023	Jun 2023	May 2023	Apr 2023	Mar 2023	Feb 2023	Jan 2023
★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★★
Dec 2022	Nov 2022	Oct 2022	Sep 2022	Aug 2022	Jul 2022	Jun 2022	May 2022	Apr 2022	Mar 2022	Feb 2022	Jan 2022
★★★★★	★★★★	★★★★	★★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★
Dec 2021	Nov 2021	Oct 2021	Sep 2021	Aug 2021	Jul 2021	Jun 2021	May 2021	Apr 2021	Mar 2021	Feb 2021	Jan 2021
★★★★	★★★	★★★★	★★★	★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★

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Research Methodology for Valuing Companies

Overview

At the heart of our valuation system is a detailed projection of a company's future cash flows, resulting from our analysts' research. Analysts create custom industry and company assumptions to feed income statement, balance sheet, and capital investment assumptions into our globally standardized, proprietary discounted cash flow, or DCF, modeling templates. We use scenario analysis, indepth competitive advantage analysis, and a variety of other analytical tools to augment this process. Moreover, we think analyzing valuation through discounted cash flows presents a better lens for viewing cyclical companies, high-growth firms, businesses with finite lives (e.g., mines), or companies expected to generate negative earnings over the next few years. That said, we don't dismiss multiples altogether but rather use them as supporting cross-checks for our DCF-based fair value estimates. We also acknowledge that DCF models offer their own challenges (including a potential proliferation of estimated inputs and the possibility that the method may miss shortterm market-price movements), but we believe these negatives are mitigated by deep analysis and our longterm approach.

Morningstar's equity research group ("we," "our") believes that a company's intrinsic worth results from the future cash flows it can generate. The Morningstar Rating for stocks identifies stocks trading at a discount or premium to their intrinsic worth—or fair value estimate, in Morningstar terminology. Five-star stocks sell for the biggest risk adjusted discount to their fair values, whereas 1-star stocks trade at premiums to their intrinsic worth.

Four key components drive the Morningstar rating: (1) our assessment of the firm's economic moat, (2) our estimate of the stock's fair value, (3) our uncertainty around that fair value estimate and (4) the current market price. This process ultimately culminates in our singlepoint star rating.

1. Economic Moat

The concept of an economic moat plays a vital role not only in our qualitative assessment of a firm's long-term investment potential, but also in the actual calculation of our fair value estimates. An economic moat is a structural feature that allows a firm to sustain excess profits over a long period of time. We define economic profits as returns on invested capital (or ROIC) over and above our estimate of a firm's cost of capital, or weighted average cost of capital (or WACC). Without a moat, profits are more susceptible to competition. We have identified five sources of economic moats: intangible assets, switching costs, network effect, cost advantage, and efficient scale.

Companies with a narrow moat are those we believe are more likely than not to achieve normalized excess returns for at least the next 10 years. Wide-moat companies are those in which we have very high confidence that excess returns will remain for 10 years, with excess returns more likely than not to remain for at least 20 years. The longer a firm generates economic profits, the higher its intrinsic value. We believe low-quality, no-moat companies will see their normalized returns gravitate toward the firm's cost of capital more quickly than companies with moats.

When considering a company's moat, we also assess whether there is a substantial threat of value destruction, stemming from risks related to ESG, industry disruption, financial health, or other idiosyncratic issues. In this context, a risk is considered potentially value destructive if its occurrence would eliminate a firm's economic profit on a cumulative or midcycle basis. If we deem the probability of occurrence sufficiently high, we would not characterize the company as possessing an economic moat.

2. Estimated Fair Value

Combining our analysts' financial forecasts with the firm's economic moat helps us assess how long returns on invested capital are likely to exceed the firm's cost of capital. Returns of firms with a wide economic moat rating are assumed to fade to the perpetuity period over a longer period of time than the returns of narrow-moat firms, and both will fade slower than no-moat firms, increasing our estimate of their intrinsic value.

Our model is divided into three distinct stages:

Stage I: Explicit Forecast

In this stage, which can last five to 10 years, analysts make full financial statement forecasts, including items such as revenue, profit margins, tax rates, changes in workingcapital accounts, and capital spending. Based on these projections, we calculate earnings before interest, after taxes (EBI) and the net new investment (NNI) to derive our annual free cash flow forecast.

Stage II: Fade

The second stage of our model is the period it will take the company's return on new invested capital—the return on capital of the next dollar invested ("RONIC")—to decline (or rise) to its cost of capital. During the Stage II period, we use a formula to approximate cash flows in lieu of explicitly modeling the income statement, balance sheet, and cash flow statement as we do in Stage I. The length of the second stage depends on the strength of the company's economic moat. We forecast this period to last anywhere from one year (for companies with no economic moat) to 10–15 years or more (for wide-moat companies). During this period, cash flows are forecast using four assumptions: an average growth rate for EBI over the period, a normalized investment rate, average return on new invested capital (RONIC), and the number of years until perpetuity, when excess returns cease. The investment rate and return on new invested capital decline until a perpetuity value is calculated. In the case of firms that do not earn their cost of capital, we assume marginal ROICs rise to the firm's cost of capital (usually attributable to less reinvestment), and we may truncate the second stage.

Stage III: Perpetuity

Once a company's marginal ROIC hits its cost of capital, we calculate a continuing value, using a standard perpetuity formula. At perpetuity, we assume that any growth or decline or investment in the business neither creates nor destroys value and that any new investment provides a return in line with estimated WACC.

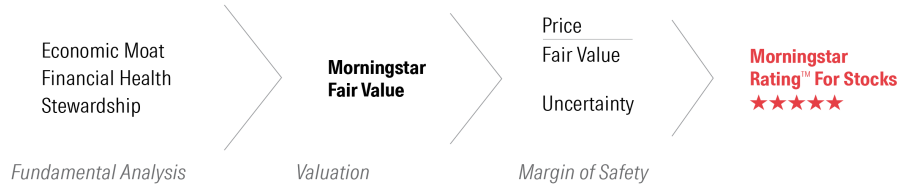
Because a dollar earned today is worth more than a dollar earned tomorrow, we discount our projections of cash flows in stages I, II, and III to arrive at a total present value of expected future cash flows. Because we are modeling free cash flow to the firm—representing cash available to provide a return to all capital providers—we discount future cash flows using the WACC, which is a weighted average of the costs of equity, debt, and preferred stock (and any other funding sources), using expected future proportionate long-term, market-value weights.

3. Uncertainty Around That Fair Value Estimate

Morningstar's Uncertainty Rating is designed to capture the range of potential outcomes for a company's intrinsic value. This rating is used to assign the margin of safety required before investing, which in turn explicitly drives our stock star rating system. The Uncertainty Rating is aimed at identifying the confidence we should have in assigning a fair value estimate for a given stock.

Our Uncertainty Rating is meant to take into account anything that can increase the potential dispersion of future

Morningstar Equity Research Star Rating Methodology



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outcomes for the intrinsic value of a company, and anything that can affect our ability to accurately predict these outcomes. The rating begins with a suggested rating produced by a quantitative process based on the trailing 12-month standard deviation of daily stock returns. An analyst overlay is then applied, with analysts using the suggested rating, historical rating data, and their own knowledge of the company to inform them as they make the final Uncertainty Rating decision. Ultimately, the rating decision rests with the analyst. Analysts take into account many characteristics when making their final decision, including cyclical factors, operational and financial factors such as leverage, company-specific events, ESG risks, and anything else that might increase the potential dispersion of future outcomes and our ability to estimate those outcomes.

Our recommended margin of safety—the discount to fair value demanded before we'd recommend buying or selling the stock—widens as our uncertainty of the estimated value of the equity increases. The more uncertain we are about the potential dispersion of outcomes, the greater the discount we require relative to our estimate of the value of the firm before we would recommend the purchase of the shares. In addition, the Uncertainty Rating provides guidance in portfolio construction based on risk tolerance.

Our Uncertainty Ratings are: Low, Medium, High, Very High, and Extreme.

	Margin of Safety	
Qualitative Analysis Uncertainty Ratings	★★★★★Rating	★Rating
Low	20% Discount	25% Premium
Medium	30% Discount	35% Premium
High	40% Discount	55% Premium
Very High	50% Discount	75% Premium
Extreme	75% Discount	300% Premium

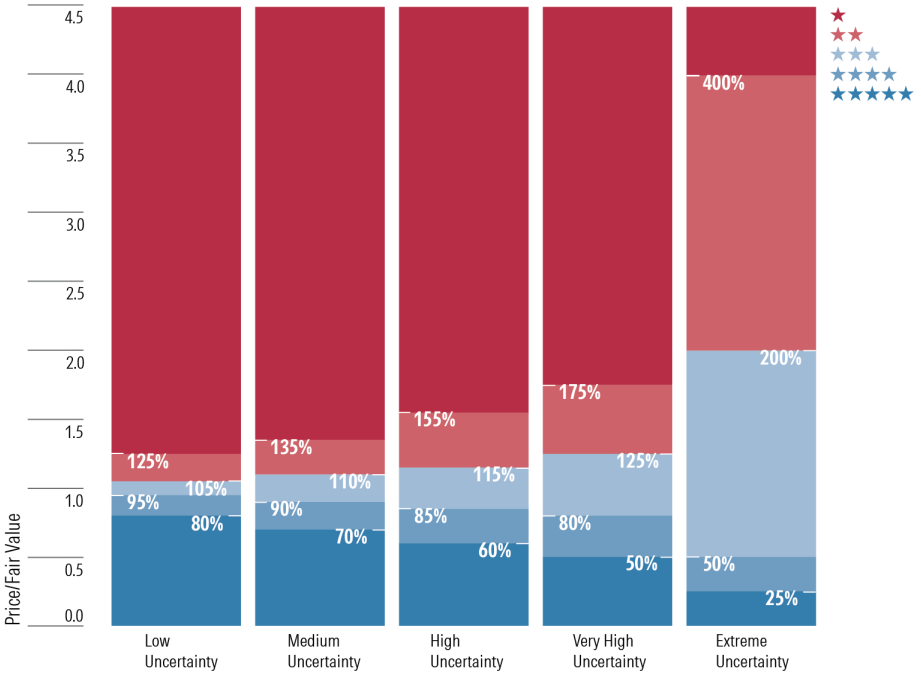
Our uncertainty rating is based on the interquartile range, or the middle 50% of potential outcomes, covering the 25th percentile–75th percentile. This means that when a stock hits 5 stars, we expect there is a 75% chance that the intrinsic value of that stock lies above the current market price. Similarly, when a stock hits 1 star, we expect there is a 75% chance that the intrinsic value of that stock lies below the current market price.

4. Market Price

The market prices used in this analysis and noted in the report come from exchange on which the stock is listed which we believe is a reliable source.

For more details about our methodology, please go to https://shareholders.morningstar.com

Morningstar Equity Research Star Rating Methodology



Morningstar Star Rating for Stocks

Once we determine the fair value estimate of a stock, we compare it with the stock's current market price on a daily basis, and the star rating is automatically re-calculated at the market close on every day the market on which the stock is listed is open. Our analysts keep close tabs on the companies they follow, and, based on thorough and ongoing analysis, raise or lower their fair value estimates as warranted.

Please note, there is no predefined distribution of stars. That is, the percentage of stocks that earn 5 stars can fluctuate daily, so the star ratings, in the aggregate, can serve as a gauge of the broader market's valuation. When there are many 5-star stocks, the stock market as a whole is more undervalued, in our opinion, than when very few companies garner our highest rating.

We expect that if our base-case assumptions are true the market price will converge on our fair value estimate over time generally within three years (although it is impossible to predict the exact time frame in which market prices may adjust).

Our star ratings are guideposts to a broad audience and individuals must consider their own specific investment goals, risk tolerance, tax situation, time horizon, income needs, and complete investment portfolio, among other factors.

The Morningstar Star Ratings for stocks are defined below:

★★★★★ We believe appreciation beyond a fair risk adjusted return is highly likely over a multiyear time frame. Scenario analysis developed by our analysts indicates that the current market price represents an excessively pessimistic outlook, limiting downside risk and maximizing upside potential.

★★★★ We believe appreciation beyond a fair risk-adjusted return is likely.

★★★ Indicates our belief that investors are likely to receive a fair risk-adjusted return (approximately cost of equity).

★★ We believe investors are likely to receive a less than fair risk-adjusted return.

★ Indicates a high probability of undesirable risk-adjusted returns from the current market price over a multiyear time frame, based on our analysis. Scenario analysis by our analysts indicates that the market is pricing in an excessively optimistic outlook, limiting upside potential and leaving the investor exposed to Capital loss.

Other Definitions

Last Price: Price of the stock as of the close of the market of the last trading day before date of the report.

Capital Allocation Rating: Our Capital Allocation (or Stewardship) Rating represents our assessment of the quality of management's capital allocation, with particular emphasis on the firm's balance sheet, investments,

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and shareholder distributions. Analysts consider companies' investment strategy and valuation, balance sheet management, and dividend and share buyback policies. Corporate governance factors are only considered if they are likely to materially impact shareholder value, though either the balance sheet, investment, or shareholder distributions. Analysts assign one of three ratings: "Exemplary", "Standard", or "Poor". Analysts judge Capital Allocation from an equity holder's perspective. Ratings are determined on a forward looking and absolute basis. The Standard rating is most common as most managers will exhibit neither exceptionally strong nor poor capital allocation.

Capital Allocation (or Stewardship) analysis published prior to Dec. 9, 2020, was determined using a different process. Beyond investment strategy, financial leverage, and dividend and share buyback policies, analysts also considered execution, compensation, related party transactions, and accounting practices in the rating.

Capital Allocation Rating: Our Capital Allocation (or Stewardship) Rating represents our assessment of the quality of management's capital allocation, with particular emphasis on the firm's balance sheet, investments, and shareholder distributions. Analysts consider companies' investment strategy and valuation, balance sheet management, and dividend and share buyback policies. Corporate governance factors are only considered if they are likely to materially impact shareholder value, though either the balance sheet, investment, or shareholder distributions. Analysts assign one of three ratings: "Exemplary", "Standard", or "Poor". Analysts judge Capital Allocation from an equity holder's perspective. Ratings are determined on a forward looking and absolute basis. The Standard rating is most common as most managers will exhibit neither exceptionally strong nor poor capital allocation.

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Sustainalytics ESG Risk Rating Assessment: The ESG Risk Rating Assessment is provided by Sustainalytics; a Morningstar company.

Sustainalytics' ESG Risk Ratings measure the degree to which company's economic value at risk is driven by environment, social and governance (ESG) factors.

Sustainalytics analyzes over 1,300 data points to assess a company's exposure to and management of ESG risks. In other words, ESG Risk Ratings measures a company's unmanaged ESG Risks represented as a quantitative score. Unmanaged Risk is measured on an open-ended scale starting at zero (no risk) with lower scores representing less unmanaged risk and, for 95% of cases, the unmanaged ESG Risk score is below 50.

Based on their quantitative scores, companies are grouped into one of five Risk Categories (negligible, low, medium, high, severe). These risk categories are absolute, meaning that a 'high risk' assessment reflects a comparable degree of unmanaged ESG risk across all subindustries covered.

The ESG Risk Rating Assessment is a visual representation of Sustainalytics ESG Risk Categories on a 1 to 5 scale. Companies with Negligible Risk = 5 Globes, Low Risk = 4, Medium Risk = 3 Globes, High Risk = 2 Globes, Severe Risk = 1 Globe. For more information, please visit sustainalytics.com/esg-ratings/

Ratings should not be used as the sole basis in evaluating a company or security. Ratings involve unknown risks and uncertainties which may cause our expectations not to occur or to differ significantly from what was expected and should not be considered an offer or solicitation to buy or sell a security.

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